

Research Internship in Mathematics

LAREMA — UMR 6093, CNRS — University of Angers

Jo-Sheun AVRIL — June 2026 — Supervisor: [Prof. Rodolphe GARBIT]

RESEARCH QUESTION

Two questions connected by a single object: power series.

(i) **Behaviour on the boundary.** Can a power series converge pointwise at *every* point of the unit circle without converging uniformly on the closed disc? Equivalently: does pointwise convergence on the boundary impose any global regularity?

(ii) **Generating functions and stochastic processes.** The generating function of a discrete law encodes it entirely and converts probabilistic questions into analytic ones — applied here to Galton–Watson branching processes and to random walks.

MATHEMATICAL TOOLS

Abel’s theorem and radial convergence; pointwise / uniform / normal convergence. **Cauchy product**, **Stirling’s formula**, asymptotic estimation of coefficients, integral comparison and harmonic-sum bounds. **Generating functions**: composition theorem for random sums, moments as radial derivatives at $t = 1$, Möbius transformations and matrix diagonalisation for iterates.

RESULTS AND PROOFS

A power series converging on the whole closed disc without converging uniformly. Starting from Lowther’s function $f(z) = \sum_{n \geq 1} \frac{1}{n^5(\alpha_n - z)}$ with $\alpha_n = 1 + i/n^3$, I proved that its Taylor expansion $\sum_{k \geq 0} a_k z^k$, with $a_k = \sum_{n \geq 1} n^{-5} \alpha_n^{-(k+1)}$ and $|a_k| \leq 2$, converges at *every* point of $\mathbb{D}$ yet not uniformly. Convergence follows by splitting the remainder $R_K(z)$ at an index n_0 : the tail is bounded by a convergent series, the head by q^{K+1} with $q = \max_{n \leq n_0} |\alpha_n|^{-1} < 1$. For unboundedness, along $z_m = \sqrt{1 - m^{-6}} + i m^{-3} \in \mathbb{U}$ the diagonal term satisfies $m \leq T_m \leq 2m$, while a three-zone estimate ($\sum n^{-2}$, a harmonic bound $8H_m/m$, an integral comparison) bounds all other terms by 11 *uniformly in m*, giving $|f(z_m)| \geq m - 11 \rightarrow \infty$. As partial sums are bounded and a uniform limit of bounded functions is bounded, convergence cannot be uniform; the radius is exactly 1.

Generalisation (my own). Replacing (5, 3) by $\alpha_n = 1 + i/n^t$ and a factor n^{-s} , I showed the construction works precisely when $t + 1 < s < 2t$ — convergence at $z = 1$ requiring $s > t + 1$, unboundedness requiring $s < 2t$ — so Lowther’s pair is simply the smallest integer solution.

A converse. I proved that if the sign of a_n depends only on $n \bmod m$, convergence at the m -th roots of unity forces absolute convergence of $\sum |a_n|$, hence *uniform* convergence on $\mathbb{D}$ — isolating why naive sign patterns cannot produce a counterexample.

Branching processes and random walks. Extinction probability as the smallest fixed point of the offspring generating function, almost sure iff $m \leq 1$; worked cases (Bernoulli, binomial, geometric — the last by diagonalising the associated Möbius matrix to obtain G_n in closed form) plus an application to nucleotide replication. For the random walk: recurrence iff $p = q = \frac{1}{2}$ via Stirling, and the first-passage law from $F_1(s) = qs + ps F_1(s)^2$, giving $\mathbb{P}(\tau_{-1} = n + 1) = C_n p^n q^{n+1}$ with C_n the Catalan numbers.

An unexpected identity. Euler’s polygon-triangulation problem yields $C(x) = 1 + xC(x)^2$ — formally the same equation, hence the same Catalan numbers. I identified the structural reason: both objects decompose into two independent sub-objects of the same nature, which is precisely what produces a quadratic term.

PERSONAL CONTRIBUTION

I wrote every proof in the report myself, from the primary literature, with my supervisor advising on structure and reviewing the arguments.

Beyond reproducing Lowther’s example, three parts are my own: the generalisation to the admissibility region $t + 1 < s < 2t$; the converse showing that periodic sign patterns force uniform convergence; and the identification of the shared quadratic structure linking first-passage times to Euler’s triangulation problem.

TRANSFERABLE SKILLS

Probability: generating functions, branching processes, random walks — the founding objects of stochastic process theory. **Analytical rigour**: constructing counterexamples and identifying precisely *which* hypothesis a result depends on; deriving bounds uniform in a parameter.

Asymptotic analysis and **research autonomy**: several weeks from primary literature to fully written proofs, including a generalisation of the published result.

RELEVANCE TO QUANTITATIVE FINANCE

This internship was a piece of pure mathematics; no financial application was treated. The objects studied are nevertheless those on which several quantitative fields are built.

Random walks are the discrete model whose limit is Brownian motion, the foundation of stochastic calculus and asset pricing. **First-passage times** formalise threshold-crossing: barrier options, structural models of default, ruin theory in actuarial science. **Branching processes** model propagation, notably default cascades in credit risk. **Generating functions** belong to the same transform family as characteristic functions, used in option pricing and risk aggregation. This internship was therefore not a training in finance, but the acquisition of the probabilistic foundations on which it rests.